

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2023-24

Paper: BVCA-402 — Probability and Statistics

Time: Three Hours

Maximum Marks: 70

Instructions: Answer the questions in each section as instructed. All questions carry marks as indicated.

SECTION — A

Long-Answer Questions. Answer any two questions:

2 × 12 = 24 Marks**Question 1.** Derive the mean and variance of the Normal distribution $N(\mu, \sigma^2)$.**Question 2.** What is a simple linear regression model? Obtain the ordinary least squares expressions for the estimators β_0 and β_1 .**Question 3.** What is a cumulative distribution function? Explain the properties of the distribution function with their mathematical proofs.**Question 4.** What is a likelihood function? Explain the theory of how to obtain the maximum likelihood estimator (MLE) of the parameter of any distribution. In random sampling from a Normal distribution $N(\mu, \sigma^2)$, find the maximum likelihood estimator (MLE) for:

- (a) μ when σ^2 is known,
- (b) σ^2 when μ is known.

SECTION — B

Question 5. Attempt any six parts (Answer in about 200 words each): **6 × 6 = 36 Marks**

- (a) Calculate the mean and standard deviation of the continuous random variable X if its probability density function $f(x)$ is given by:

$$f(x) = \begin{cases} \frac{3}{56}x^2 & \text{if } 2 < x < 4 \\ 0 & \text{otherwise} \end{cases}$$

- (b) From a city population, the probability of selecting:
 - (i) a male or a smoker is $7/10$,
 - (ii) a male smoker is $2/5$,
 - (iii) a male, if a smoker is already selected, is $2/3$.

Find the probability of selecting:

- (i) a non-smoker,

- (ii) a male,
 - (iii) a smoker, if a male is first selected.
- (c) The contents of three urns I, II, and III are as follows:
- Urn I: 1 white, 2 black, and 3 red balls
 - Urn II: 2 white, 1 black, and 1 red balls
 - Urn III: 4 white, 5 black, and 3 red balls
- One urn is chosen at random and two balls are drawn from it. They happen to be white and red. What is the probability that they come from:
- (i) Urn I,
 - (ii) Urn II,
 - (iii) Urn III?
- (d) Define the chi-square test statistic. What are the essential conditions for the validity of the chi-square test?
- (e) What are non-parametric tests? Define the Kolmogorov-Smirnov statistic.
- (f) For any three events A , B , and C , prove the following:
- (i) $P(A \cup B | C) = P(A | C) + P(B | C) - P(A \cap B | C)$
 - (ii) $P(B | A) \leq P(C | A)$ provided that $B \subseteq C$ and $P(A) > 0$.
- (g) Discuss statistical hypothesis. Define null and alternative hypothesis with an illustrative example.
- (h) State and define Chebyshev's inequality and Markov's inequality.
- (i) Define Statistics. According to you, where can it be applied? Give at least three practical examples.

SECTION — C

Question 6. Attempt all parts:

$10 \times 1 = 10$ Marks

- (a) In an experiment, a coin is tossed five times. Write down the sample space.
- (b) State the additive property of the Poisson distribution.
- (c) Define the Weak Law of Large Numbers (WLLN).
- (d) Define Type-II error in hypothesis testing.
- (e) Who discovered the Normal distribution?
- (f) Write down the probability density function (PDF) of a standard normal variate.
- (g) Give two practical examples of a Poisson distribution.

- (h) For any three events A , B , and C , write the expression for $P(A \cup B \cup C)$.
- (i) Define exhaustive events.
- (j) Write down the mean and variance of a standard normal variate.